

Frame Dependence: Circuit Robustness Under Task-Preserving Distribution Shift

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DRAFT — 59 unmeasured values remain.

Every red bracket marks a number the experiment has not yet returned. No value in this document is estimated, extrapolated or illustrative: a number is either measured or visibly absent.

Abstract

This paper measures whether a circuit discovered on a diverse set of sentence frames remains faithful on frames it was not discovered on. Across **[PENDING: n]** task families—indirect object identification, factual recall, subject–verb agreement and in-context induction—on Llama-3.2-3B and Pythia-1.4B, we find **[PENDING: headline: does diverse discovery restore transfer, and does it depend on family]. [PENDING: secondary: frame axis versus entity axis]**.

The question is open because published results conflict. Merullo et al. report that the GPT-2 Small indirect-object circuit “generalizes surprisingly well” to prompt variants designed to break it. Rai et al. report cross-dataset faithfulness drops reaching 79 percentage points under variations that preserve task semantics, and conclude that circuit discovery recovers dataset-specific implementations. A third study finds near-zero circuit overlap between template groups. One candidate reconciliation is that the outcome is set by the *discovery* data: a circuit found on a narrow set has no opportunity to encode anything but that set—with a single sentence frame, token position is perfectly confounded with syntactic role—while the circuit Merullo et al. evaluate came from a fifteen-template protocol. If that is the explanation, the remedy is a better discovery set rather than a new discovery algorithm, which is what the field has so far proposed.

Deciding it requires a shift design that prior work does not supply: one holding the number of templates fixed while changing only their identity. Comparing a one-frame discovery set against a fifteen-frame evaluation set moves count and identity together and cannot separate them. We discover a circuit on D and score it on D' , where both contain the same number of templates and share none, report the ratio $F(C_D | D')/F(C_D | D)$ as *retention*, and admit a result only where the model still performs the task on D' . Frame shift and entity shift are reported on separate axes rather than merged into a single held-out number.

We release the protocol, twelve matched-diversity dataset pairs and reference implementations, all runnable on a single consumer machine.

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1 Introduction

Mechanistic interpretability is growing quickly and validating slowly. A survey of indexed publications (§2.2) finds [PENDING: N] papers matching “mechanistic interpretability”, [PENDING: pct]% of them from the last two years, against [PENDING: N] matching “circuit faithfulness” and [PENDING: N] matching “circuit generalization”. Circuits are being discovered far faster than they are being tested, and the standard test—faithfulness—is normally computed on the data the circuit was found on.

That is not a criticism of the metric. Faithfulness answers a well-posed question: does this subgraph, on its own, reproduce the behaviour? The problem is what readers take it to mean. A circuit reported as 87% faithful is read as a claim about how the model does the task, not about how the model does the task *on these prompts*, and nothing in the usual reporting distinguishes those.

1.1 Two Explanations With Opposite Consequences

Our own prior work [2] measured faithfulness falling from 0.663 to 0.228 (Llama-3.2-3B) and 0.854 to -0.028 (Pythia-1.4B) when circuits discovered on one sentence frame were re-scored on fifteen, while both models continued to perform the task. We reported the mechanism as unverified, and it is worth being precise about why that caveat was unavoidable. The comparison moved two variables together:

	discovery set	evaluation set
templates	1	15
identity	frame f_1	frames $f_1 \dots f_{15}$

Two stories fit that evidence equally well.

(a) Circuits are frame-specific. A circuit found on any frame fails on other frames. If so, a large and accelerating literature is reporting numbers that overstate what has been established, and the remedy is procedural: faithfulness must be reported on held-out data, always.

(b) The discovery set was degenerate. With a single frame, token position is perfectly confounded with syntactic role. A head that appears circuit-critical may be responding to a fixed position, and such a head fails anywhere the coincidence breaks. If so, the published collapse is a fact about single-template discovery rather than about circuits, and the remedy is upstream and cheap: discover on diverse data.

These prescribe opposite things—one says change how you report, the other says change how you search—and no comparison that varies template count can choose between them.

1.2 Contributions

Prior work has established that circuits often fail to transfer and that frame and lexical shift are distinct axes; §?? states precisely which of our earlier claims those results anticipate, and we do not restate them as new. What this paper adds:

1. **A test of the cheap remedy.** The field’s response to dataset-specific circuits has been to propose new discovery algorithms. We test whether simply discovering on a diverse set is sufficient, which is the intervention a practitioner would try first and which no prior study isolates (§3).
2. **Matched-diversity disjoint splits.** Template count held fixed, template identity disjoint—the design that separates “the frames differ” from “there are more frames”. Prior variation axes are systematic transformations or lexical substitutions and do not make this separation (§3.3).
3. **Retention as the reported quantity.** $F(C_D | D')/F(C_D | D)$ rather than $F(C_D | D')$, which distinguishes a circuit that was weak everywhere from one that collapsed off its discovery set (§3).
4. **Measurements** across **[PENDING: n]** families and two models, one of them a family not covered at this scale by prior work (§6).
5. **Two harness defects, disclosed** (§5.2), both silent on near-uniform-length data and both capable of contaminating this comparison.

2 Background

2.1 Circuits and Faithfulness

Following Wang et al. [5], a circuit C is a set of model components—here attention heads—hypothesised to implement a behaviour. Faithfulness mean-ablates every component outside C and measures the recovered logit difference, normalised against the fully-ablated floor:

$$F(C) = \frac{\text{LD}(\text{only } C \text{ active}) - \text{LD}_{\text{floor}}}{\text{LD}(\text{clean}) - \text{LD}_{\text{floor}}}$$

The floor term matters. An unnormalised ratio implicitly treats the fully-ablated model as scoring zero, which is not generally true and can be negative.

2.2 Field Context

Validation of circuits has not kept pace with their discovery. Indexed publication counts (OpenAlex, exact-phrase title-and-abstract search, retrieved 2026-08-13) place “mechanistic interpretability” at 2,957 works, 87% of them from 2025–2026, and “activation patching” at 561. Terms naming the validation question are two orders of magnitude rarer.

Table 1: Exact-phrase publication counts, OpenAlex, 2026-08-13. Counts use quoted phrase search; the default fuzzy search is not phrase search and inflates `activation patching` from 561 to 41,006. Terms whose usage is dominated by electrical-circuit senses (`circuit validation`, `circuit robustness`) are excluded.

phrase	all	2025–26	recent
mechanistic interpretability	2,957	2,578	87%
activation patching	561	442	79%
circuit generalization	8	1	12%
circuit faithfulness	8	5	62%

These counts measure vocabulary rather than activity and should be read only as an order of magnitude. The three studies most directly relevant to this paper (§2.3) use none of the phrases above, describing the same problem as adaptive circuit behaviour, dataset-specific circuits and rephrasing variance. The literature review in §2.3, not Table 1, establishes what is known.

2.3 Related Work: The Field Disagrees With Itself

Circuit robustness is not an unexamined question. It is an examined question with contradictory answers, and that contradiction is what this paper is positioned against.

Evidence that circuits generalize. Merullo et al. [3] test the GPT-2 Small IOI circuit on prompt variants that violate the assumptions of the published algorithm, and report that it “generalizes surprisingly well, reusing all of its components and mechanisms while only adding additional input edges”. They identify a previously undescribed mechanism, S2 Hacking, that accounts for the circuit succeeding on formats where the original algorithm should fail, and conclude that circuits may be more flexible than previously recognised.

Evidence that circuits do not generalize. Rai et al. [4] test the single-circuit-per-task assumption across four tasks (indirect object identification, entity binding, arithmetic addition, sequence completion) on GPT-2 Small, Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, varying datasets along complexity, syntax and domain axes while preserving task semantics. They report in-distribution to cross-dataset faithfulness drops reaching 79 percentage points at circuit size 0.05, and conclude that existing methods recover dataset-specific rather than task-general circuits. Independently, Authors of arXiv:2606.16920 [1] study rephrasing variance and find that different templates activate substantially different circuits, with near-zero overlap between template groups, across GPT-2 and Pythia models on subject–verb agreement, IOI, greater-than and coding tasks.

These cannot all be straightforwardly true. One candidate reconciliation is that the answer depends on the discovery set. Merullo et al. [3] evaluate a circuit produced by the Wang et al. protocol, which uses fifteen templates with balanced name ordering; a circuit discovered on a narrow set has no opportunity to encode anything but the narrow set. If that is the reconciliation, frame dependence is a property of *how the circuit was found* rather than of circuits as such, and the remedy is cheaper than it currently appears.

3 The Protocol

3.1 Definitions

Let $\mathcal{T}$ be a task, and let D and D' be two datasets that both instantiate $\mathcal{T}$. Discover C_D on D by any procedure; here, the combined patching-and-ablation ranking of §5.1. Then report:

$$\text{retention } R = \frac{F(C_D|D')}{F(C_D|D)} \quad \text{frame dependence } \Delta = F(C_D|D) - F(C_D|D')$$

R is the headline because it is scale-free; Δ is reported alongside because R is unstable when $F(C_D|D)$ is near zero.

3.2 Matched Diversity

D and D' must have the same number of templates and share none. This is the whole design. Holding count fixed removes the confound of §1; sharing no frames is what makes the shift a shift.

We additionally hold clause *shape* constant across the two template banks where the family permits it. Varying syntax as well would broaden the shift, but it would also reintroduce a second moving part, and the point of this protocol is that exactly one thing moves.

3.3 Shift Axes

axis	templates	entity pool
frame	disjoint	shared
entity	shared	disjoint
both	disjoint	disjoint

Reporting only a merged “held-out” number would conflate two different objects. A circuit that survives new names but not new frames is a syntactic-position circuit; a circuit that survives new frames but not new names has memorised lexical items. Both are interesting; they are not the same finding.

3.4 The Task-Preservation Gate

A faithfulness collapse on D' means nothing about circuits if the model cannot do the task on D' . We therefore report clean logit difference on both splits and admit a row only if behaviour is retained above a threshold of 0.8; failing rows are listed as excluded, with their numbers, rather than dropped silently or averaged in.

4 Datasets

Four families, each in three shift axes, each split into D and D' of 200 examples:

family	mechanism exercised	templates/bank	answer contrast
ioi	routing / name suppression	15	indirect object vs subject
factual	retrieval	6	capital vs other capital
agreement	long-range syntactic dependency	6	<i>are</i> vs <i>is</i>
induction	in-context copying	6	paired vs unpaired token

Bank A for **ioi** reproduces the fifteen frames of Melton [2] exactly, so D matches the previously published evaluation set and the new D' is the only novel object.

4.1 Construction Constraints

Three constraints are enforced at generation time; each was added because violating it produces a plausible-looking but wrong measurement rather than an error.

Single-token answers. The metric reads one token id at the final position, so a multi-token answer would be scored on its first piece—often a shared prefix, and therefore not diagnostic. Candidate answers are filtered to single tokens under *both* tokenizers.

Clean/corrupt token-length parity. Activation patching caches per-head output on a clean pass and splices it into a corrupt pass at the same tensor positions. That correspondence requires position i to mean the same thing in both. IOI corruption swaps two single-token names and is length-preserving by construction; entity substitution in other families is not. Pairs failing parity under either tokenizer are dropped—**[PENDING: pct]**% of **factual** candidates, and none elsewhere.

Template-balanced sampling. The parity filter does not drop uniformly across frames, so taking the first n survivors would skew the template mix and confound frame identity with frame frequency. Records are taken round-robin across template ids.

Entity pools are filtered before splitting, not after. Splitting into fixed banks and filtering each independently lets tokenizer attrition fall unevenly; in an early version this left one induction bank with five usable words and the other with one. An unequal pool is itself a difference between D and D' .

5 Method

5.1 Discovery

Two per-head metrics over a full layers $\times$ heads sweep:

Patching. Run the corrupt prompt with head (l, h) ’s clean output spliced in, and measure the fraction of the clean-corrupt gap recovered. High means the head carries task signal.

Ablation. Mean-ablate (l, h) on the clean prompt and measure the logit-difference drop. High means nothing else compensates for its loss.

Each is min-max normalised across all heads and summed; the circuit is the top ten by that sum. The two metrics disagree often enough to matter, which is why the selection uses both.

5.2 Implementation Requirements

Two implementation properties are required for this comparison and are not guaranteed by a harness built for a single near-uniform dataset. Both are silent when violated: they produce numbers rather than errors.

The mean-ablation reference must be computed over the full example set. A caching step that assigns rather than accumulates leaves only the final batch’s mean, so the ablation value is estimated from a fraction of the data. The quantity estimated is the same, but its variance is not, and D and D' must be scored against the same reference for their difference to mean anything.

Sequences must be padded to one global length, not per batch. The cached mean carries a sequence dimension, so a mean cached at one padded length cannot be broadcast onto a batch padded to another. Datasets of near-uniform length hide this; three of the four families here vary enough in length to expose it.

The harness used for all measurements below satisfies both. Figures produced under these corrections do not reproduce previously published IOI values to the last decimal, and the two are not presented in a common column.

6 Results

Table 2: Retention under frame shift. $R = F(C_D | D')/F(C_D | D)$. “behav.” is clean logit difference on D' as a fraction of D ; rows below 0.8 fail the task-preservation gate and are excluded from any aggregate.

model	family	$F(C_D D)$	$F(C_D D')$	R	behav.	
Llama-3.2-3B	ioi	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Llama-3.2-3B	factual	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Llama-3.2-3B	agreement	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Llama-3.2-3B	induction	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Pythia-1.4B	ioi	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Pythia-1.4B	factual	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Pythia-1.4B	agreement	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Pythia-1.4B	induction	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]

Table 3: Frame versus entity shift, IOI only.

model	R (frame)	R (entity)	R (both)	$F(C_D D)$
Llama-3.2-3B	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]
Pythia-1.4B	[PENDING:]	[PENDING:]	[PENDING:]	[PENDING:]

6.1 Circuit Overlap

[PENDING: How much do C_D and a circuit discovered directly on D' overlap? A circuit that fails to transfer but whose replacement shares most of its heads is a different phenomenon from one replaced wholesale.]

7 Analysis

Drafting note — to be written once Table 2 is filled, and to be written once. The measurement selects which of these the paper argues; it is not a menu to hedge across. Both branches share the framing above, so no section written so far needs revisiting either way.

If retention is high (R near 1 across families): the published collapse was about single-template discovery, not about circuits. The prescription is upstream and cheap, the finding bounds our own prior paper, and the protocol’s value is as a cheap check that a discovery set was not degenerate.

If retention is low: frame dependence is real, general across mechanisms as different as retrieval and induction, and faithfulness reported in-distribution systematically overstates what has been shown.

If retention splits by family: the most informative outcome, and the one that would need the most careful writing—which mechanisms are portable and which are frame-bound is a stronger claim than either uniform result.

8 Limitations

Two models, both small. Llama-3.2-3B and Pythia-1.4B. Whether frame dependence changes with scale is untested here.

Attention heads only. MLP components are not in the circuit vocabulary, so a circuit that appears to lose faithfulness may have had its work taken over by components this method cannot see.

One discovery procedure. Retention is a property of the circuit-finding method as much as of the model. A different procedure may produce more portable circuits, and this protocol is the way to check that rather than an argument that it would not.

Templates are hand-written. Frame banks were authored, not sampled from a corpus, so “frame shift” means the shift the author constructed.

The shift is synthetic. D' is task-preserving by construction and gated on behaviour, but it is not natural text.

9 Reproducibility

Code and data: <https://github.com/american-code/ResearchPapers>. Archived at **[PENDING: Zenodo concept DOI]**.

artifact	path
dataset generator	data/circuit-robustness/generate_shift_pairs.py
intervention library	data/circuit-robustness/circuit_lib.py
discovery	data/circuit-robustness/discover_circuit.py
scoring	data/circuit-robustness/score_faithfulness.py
end-to-end protocol	data/circuit-robustness/run_protocol.sh
generated splits	data/circuit-robustness/*-{D,Dprime}.json
results	data/circuit-robustness/results/

All runs are seeded (42) and executed on a single Apple M1 Max with 32 GB unified memory via MLX.

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